CS2201 Computer Organization & Architecture

L22: Basic Cache Optimization Techniques



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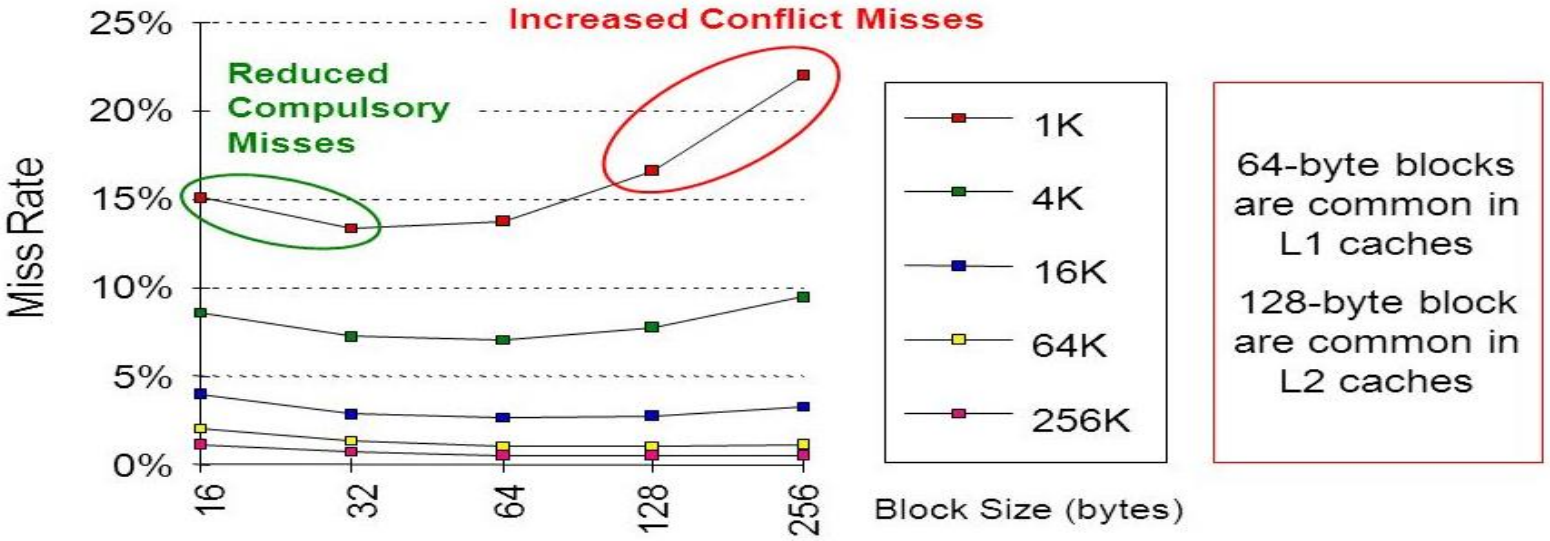
How to optimize cache ?

- Reduce Average Memory Access Time
- AMAT= Hit Time + Miss Rate x Miss Penalty
- Motives
  - Reducing the miss rate
  - Reducing the miss penalty
  - Reducing the hit time

Larger Block Size

- **Larger block size to reduce miss rate**
- **Advantages**
  - Utilize spatial locality
  - Reduces compulsory misses
- **Disadvantages**
  - Increases miss penalty
  - More time to fetch a block to the cache [bus width issue]
  - Increases conflict misses
  - More number of blocks will be mapped to the same location
  - May bring useless data and evict useful data [pollution]

Larger Block Size



Larger Caches

- **Larger cache to reduce miss rate**
- **Advantages**
  - Reduces capacity misses
  - Can accommodate larger memory footprint

	Cache size			
Block size	4K	16K	64K	256K
16	8.57%	3.94%	2.04%	1.09%
32	7.24%	2.87%	1.35%	0.70%
64	7.00%	2.64%	1.06%	0.51%
128	7.78%	2.77%	1.02%	0.49%
256	9.51%	3.29%	1.15%	0.49%

- **Drawbacks**
  - Longer hit time
  - Higher cost, area and power

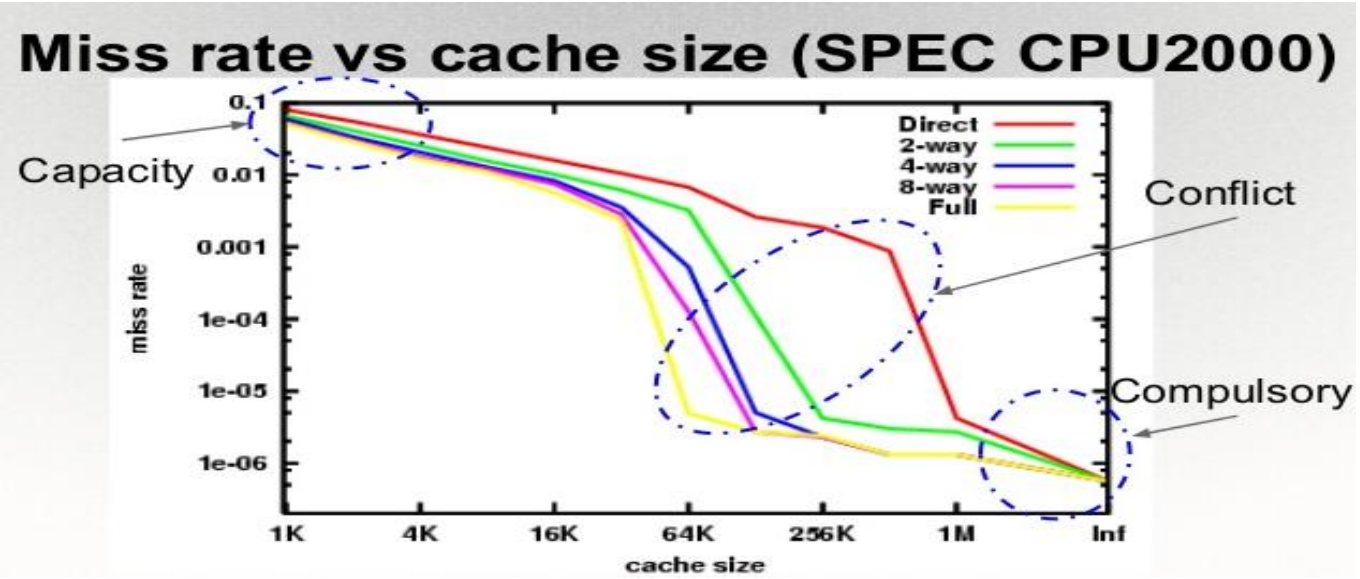
Larger Caches



Higher Associativity

- **Higher associativity to reduce miss rate**
  - Fully associative caches are the best, but high hit time.
  - So increase the associativity to an optimal possible level
- **Advantages**
  - Reduce conflict miss
  - Reduce miss rate and eviction rate
- **Drawbacks**
  - Increase in the hit time
  - Complex design than direct mapped
  - More time to search in the set (tag comparison time)

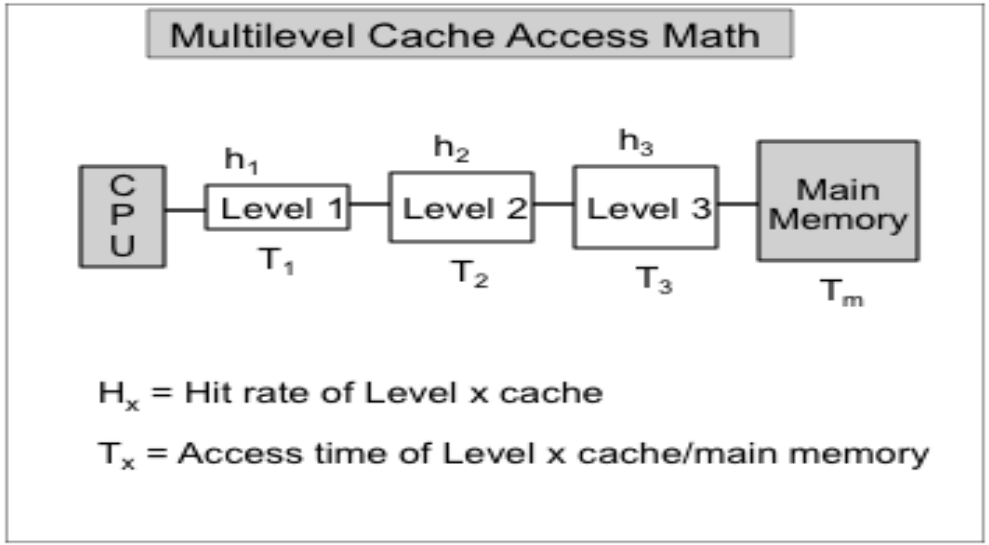
AMAT vs cache associativity



Multilevel caches

- **Multilevel caches to reduce miss penalty**
- **Caches should be faster** to keep pace with the speed of processors, **AND cache should be larger** to overcome the widening gap between the processor and main memory
- Add another level of cache between the cache and memory.
- The first-level cache (L1) can be small enough to match the clock cycle time of the fast processor. [Low hit time]
- The second-level cache (L2) can be large enough to capture many accesses that would go to main memory, thereby lessening the effective miss penalty. [Low miss rate]

Multilevel caches



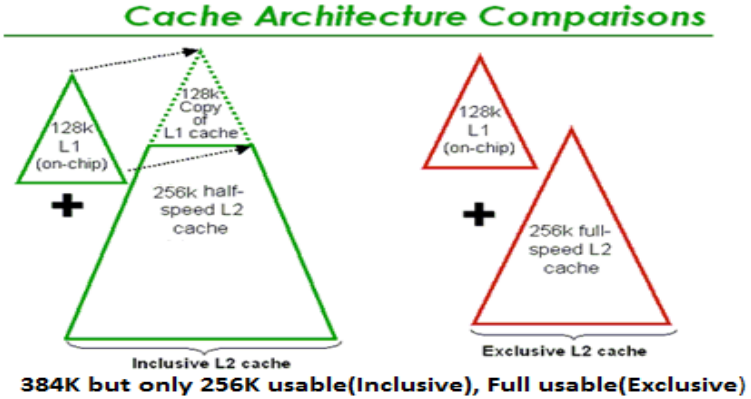
$$\text{Average memory access time} = \text{Hit time}_{L1} + \text{Miss rate}_{L1} \times \text{Miss penalty}_{L1}$$

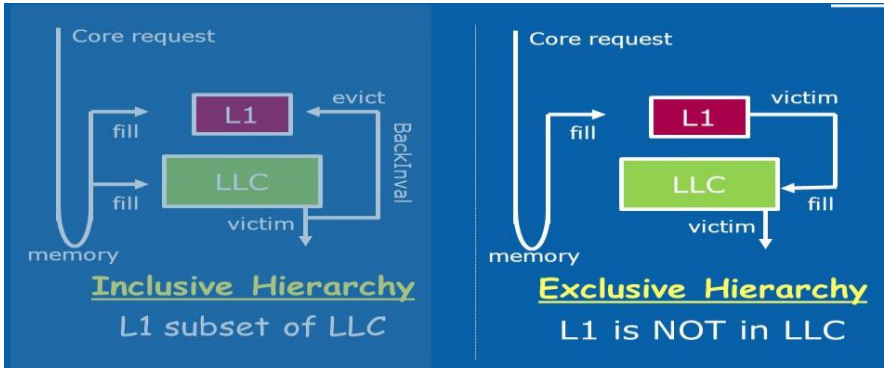
$$\text{Miss penalty}_{L1} = \text{Hit time}_{L2} + \text{Miss rate}_{L2} \times \text{Miss penalty}_{L2}$$

$$\text{Average memory access time} = \text{Hit time}_{L1} + \text{Miss rate}_{L1} \times (\text{Hit time}_{L2} + \text{Miss rate}_{L2} \times \text{Miss penalty}_{L2})$$

Multilevel caches

- **Multilevel caches to reduce miss penalty**
- **Local miss rate:** Number of misses in a cache level divided by number of memory access to this level.
- **Global miss rate:** Number of misses in a cache level divided by number of memory access generated by the CPU.
- **Inclusive and Exclusive caches**





MPKI – Miss rate Relation

The MPKI of a two programs A and B are 44 and 35 respectively. If 35% of A is data access instruction, what is the data access pattern of B if both A and B have same cache hit rate?

MPKI – Miss rate Relation

MPKI-A = 44 ; MPKI-B=35

$HR_A = HR_B \rightarrow MR_A = MR_B$

MR = misses/ memory access

(misses/ instruction) / (memory access / instruction)

MR= MPI/MAPI

Given that $MR_A = MR_B$:→ (44/1000)/1.35 = (35/1000)/ MAPI-B

MAPI-B= (0.035x1.35)/0.044 = 1.073

7.3% of instructions in B are data access instructions



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